

# Submission Summary

**Conference Name**

2026 24th National Power System Conference (NPSC)

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**Track Name**

T8: Grid Flexibility and Resiliency

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**Paper ID**

64

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**Paper Title**

Flexibility-Oriented Scheduling of Thermal and Pumped Hydro Units Using Demand Envelope Characterization Under Renewable Uncertainty

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**Abstract**

The increasing penetration of renewable energy sources introduces significant variability and uncertainty into power system operations, challenging the ability of conventional thermal units to maintain supply–demand balance. This paper proposes a flexibility-oriented scheduling framework that integrates demand envelope characterization with stochastic optimization of thermal generators and pumped storage hydro (PSH) units. First, a logistic demand envelope is extracted from five years of historical Great Britain load data to define upper and lower flexibility boundaries. Second, a scenario-based unit commitment and economic dispatch model is formulated, capturing generator ramp limits, startup costs, and PSH state-of-charge dynamics. Numerical results on a realistic system with 18 thermal units and 6 PSH units over a 12-hour horizon demonstrate that PSH significantly reduces flexibility shortages, but load shedding remains necessary under high-demand scenarios due to binding ramp constraints. The proposed framework provides grid operators with a practical tool for flexibility adequacy assessment and operational planning under renewable uncertainty.

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**Submission Files**

Aghajani\_Faraz\_Conference2.pdf (4.6 Mb, 2026-05-19 13:35:30 GMT+3:30)

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**Submission Questions Response****1. Novelty of Paper**

*The manuscript strictly abides by IEEE plagiarism policy and the similarity score in % is: 11%. Flexibility-oriented stochastic UC/ED with logistic demand envelope and PSH scheduling.*

**2. Formating**

*I declare that the manuscirpt is organized and prepared as per IEEE template (<https://www.ieee.org/conferences/publishing/templates>)*

Agreement accepted

**3. Page Length**

*I declare that the paper does not exceed the allowed page limit of six (06) pages.*

Agreement accepted

**4. Author Details**

*I Declare that all authors names and details have been included in this submission in same sequence as in the manuscript.*

Agreement accepted

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